

3-Inch 4S FPV Drone

A staged plan for designing, printing, soldering, configuring, testing, and flying your first custom mini quadcopter.

Airframe	3-inch replaceable 3D-printed prototype frame
Power system	4S LiPo; 1404 motors; 3,500-4,000KV
Electronics	25.5 x 25.5mm AIO flight controller + 4-in-1 ESC
Control link	ExpressLRS 2.4GHz
Video	Analog 5.8GHz FPV
Firmware	Betaflight
Weight objective	Under 250g including battery

Mission

Build a safe, repairable first FPV aircraft while documenting the engineering decisions as a portfolio project.

Prepared for Jakota Parker

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Core rule: Do not install propellers during soldering, firmware configuration, receiver setup, or motor testing.

Build Roadmap

Complete each gate before moving forward. This prevents incompatible purchases and makes troubleshooting manageable.

1. Define	Architecture and project folder	Target specification recorded
2. Control gear	Radio, goggles, simulator	Radio works in simulator
3. Electronics	Compatibility-checked bill of materials	All dimensions and voltage ratings match
4. Frame	Printed fit-check prototype	Props and hardware clear the frame
5. Assembly	Dry fit and wiring map	Every component has a secure location
6. Solder	Electronics connected	No short; smoke-stopper test passes
7. Configure	ELRS and Betaflight operational	Controls, OSD, failsafe verified
8. Bench test	Correct motor map and direction	All tests pass with props removed
9. Flight	Controlled hover and inspection	No heat, cracks, or control problems

Project documentation structure

- 01-requirements: goals, weight limit, selected architecture
- 02-frame: STL, CAD, slicer profile, revisions
- 03-electronics: datasheets, BOM, wiring diagram
- 04-betaflight: backups, screenshots, configuration notes
- 05-build-log: dated photographs and assembly notes
- 06-test-results: weights, hover results, failures, improvements

Portfolio habit Record why you chose each part, not only what you purchased. Engineering judgment is part of the project.

Component Plan and Compatibility Rules

Recommended architecture: separate AIO FC/ESC, ELRS receiver, analog VTX, and camera. Most true 5-in-1 boards are intended for low-power 1S whoops and should not be assumed suitable for this 4S three-inch build.

AIO FC/ESC	F405; 25.5 x 25.5mm mount; 3-6S; 25A minimum, 35-40A preferred	1
Motors	1404; about 3,500-4,000KV for 4S; verify mounting and shaft	4 + optional spare
Receiver	Serial ExpressLRS 2.4GHz nano receiver	1
Camera	Analog nano camera; typically 14 x 14mm	1
VTX	Analog 5.8GHz; switchable 25-400mW	1
VTX antenna	Correct U.FL/MMCX connector; secured away from props	1
Propellers	3-inch tri-blade; correct shaft/fastening type	6+ sets
Batteries	4S 650mAh initially; XT30	3-5
Power lead	XT30 silicone-wire pigtail	1 + spare
Capacitor	Low-ESR, approximately 470uF, 25V	1
Buzzer	Small active or self-powered FPV buzzer	1
Hardware	M2 screws, nuts, standoffs, washers, straps	Assortment

Purchase order

1	RadioMaster Pocket ELRS, two 18650 cells, analog goggles, simulator, smoke stopper, LiPo storage, multimeter	Practice and prepare before the airframe exists
2	AIO, motors, ELRS receiver, camera, VTX, antenna, props, XT30, capacitor, buzzer, M2 hardware	Buy only after frame dimensions are locked
3	4S batteries, quality balance charger, spare props, spare motor	Finalize after power-system compatibility is confirmed

Frame Design and Printing

Print a dimensional mock-up first. PLA+ is acceptable for fit checking and gentle early testing, but a fully printed frame will be less rigid and crash-resistant than carbon fiber.

Wheelbase	Approximately 130-145mm diagonal motor-to-motor
Prop clearance	Full 3-inch prop circles must not overlap frame, wires, or antenna
Motor mounting	Match exact motor hole pattern; commonly 9 x 9mm or 12 x 12mm
Electronics mounting	25.5 x 25.5mm AIO pattern using M2 soft mounts
Camera	14 x 14mm nano mount with protected lens and adjustable angle
Battery	Top mount near center of gravity; strap slots with rounded edges
Serviceability	Accessible USB port; replaceable/quick-to-reprint main structure

Starting slicer settings

Material	PLA+ for mock-up; PETG, nylon, or tougher material for flight version
Layer height	0.20mm
Walls	5-7
Top/bottom	6-8 layers
Infill	80-100%; gyroid or cubic
Orientation	Arms flat in XY plane
Adhesion	Clean glass first; add 5-8mm brim for long arms; glue only if needed

Printer check

Watch the complete first layer. Stop if the nozzle scrapes glass, filament does not flow, or lines remain round and separate. The first layer should be slightly flattened and joined.

Frame gate checklist

- Motor holes match actual motors
- AIO holes match actual board
- No motor screw can touch windings
- Three-inch prop circles clear the frame
- USB port remains accessible
- Camera and antennas are protected
- Battery sits close to the center of gravity

Dry Fit and Wiring Plan

Do not solder until every component has a planned location and wire route. Photograph the dry fit before disassembling it.

- ■ Temporarily mount all four motors
- ■ Place AIO on soft M2 gummies with orientation arrow noted
- ■ Position camera and confirm field of view
- ■ Separate receiver antenna from VTX antenna and high-current wiring
- ■ Place VTX where it receives airflow
- ■ Route XT30 lead away from propellers
- ■ Place battery and check balance
- ■ Install props only for clearance, then remove them

Power and signal layout

Battery power	4S battery -> XT30 -> capacitor/AIO battery pads
Motors	AIO motor outputs M1-M4 -> three wires per motor
ELRS receiver	5V, GND, receiver TX -> FC RX, receiver RX -> FC TX
Video camera	Camera power/GND; camera video -> FC video-in/CAM
VTX	Regulated power/GND as specified; FC video-out -> VTX video; control wire -> supported UART/pad
Buzzer	FC buzzer positive/negative pads according to board manual

UART rule

For serial devices, TX and RX cross: device TX connects to flight-controller RX, and device RX connects to flight-controller TX.

Power warning

Use the exact manual for the selected AIO, camera, and VTX. Never assume every 5V, 9V, battery-voltage, camera, or video pad behaves like one from another board.

Soldering and First Power-Up

Required bench equipment: flux, suitable electronics solder, silicone wire, heat shrink, brass tip cleaner, tweezers, magnification, multimeter, PCB holder, heat-resistant mat, fume extraction, and a smoke stopper.

Absolute rule

Never solder with a LiPo connected. Never connect a freshly soldered aircraft directly to a battery for its first test.

Soldering order

1	Practice on a spare board; form shiny, compact joints	No bridges or loose strands
2	Tin XT30 pads, leads, and capacitor legs	Correct polarity confirmed
3	Install capacitor and XT30 pigtail	Multimeter shows no hard short
4	First battery power through smoke stopper	Board boots normally
5	Disconnect battery; solder motor wires	Visual inspection passes
6	Smoke-stopper power test again	No trip or excessive heat
7	Solder receiver, camera, VTX, and buzzer one system at a time	Test after each subsystem
8	Inspect under magnification; add strain relief and heat shrink	No exposed conductors or weak joints

Pre-power checklist

- ■ Propellers removed
- ■ Battery polarity checked twice
- ■ Capacitor polarity correct
- ■ No solder bridges
- ■ No loose wire strands
- ■ Positive-to-ground check completed
- ■ Antennas connected before powering VTX
- ■ Smoke stopper installed

Radio, ExpressLRS, and Betaflight

Use a unique ExpressLRS binding phrase on both the transmitter and receiver. In Betaflight, a serial ELRS receiver normally uses Serial RX on its assigned UART and CRSF as the receiver protocol.

Configuration sequence

- ■ Install/open the current Betaflight application
- ■ Connect by USB with props removed
- ■ Save a configuration backup before changes
- ■ Confirm the 3D model follows the aircraft movement correctly
- ■ Set board alignment if the AIO is rotated
- ■ Enable Serial RX on the receiver UART
- ■ Select serial receiver and CRSF
- ■ Verify roll, pitch, yaw, and throttle direction
- ■ Verify endpoints are approximately 1000 / 1500 / 2000
- ■ Assign ARM and BEEPER switches
- ■ Configure OSD: voltage, average cell voltage, flight time, link quality, warnings
- ■ Configure VTX table and legal power/channel
- ■ Verify battery-voltage reading
- ■ Configure flight-controller failsafe to stop the aircraft on link loss

Safety behavior

Test failsafe with propellers removed: arm under controlled bench conditions, switch off the transmitter, and confirm the flight controller detects link loss and stops motor outputs.

Do not tune yet

Begin with the AIO manufacturer defaults or a documented preset appropriate for the build. Do not change PIDs and filters randomly before establishing a safe hover and collecting evidence.

Motor Test, Final Assembly, and First Flight

Motor testing is performed with every propeller removed and the frame secured.

Motor test

- ■ Use Betaflight Motors tab only after acknowledging the warning
- ■ Spin one motor at a time at very low power
- ■ Verify motor number and physical location
- ■ Verify configured rotation direction
- ■ Correct direction with the supported ESC configurator
- ■ Listen for grinding or intermittent operation
- ■ Stop if a motor or ESC heats rapidly

Final assembly

- ■ Confirm props-in or props-out configuration
- ■ Install each prop with leading edge facing the configured rotation
- ■ Secure all wires and antennas away from props
- ■ Add XT30 strain relief
- ■ Tighten screws without contacting motor windings
- ■ Weigh aircraft with battery and record result
- ■ Inspect printed frame for layer separation

First-flight procedure

Site	Open field; clear of people, animals, vehicles, structures, and obstacles
Before power	Radio on; correct model selected; goggles on; antennas installed
Power	Connect full battery; confirm video, OSD, receiver link, and warnings
Hover	Arm from behind; hover 2-3 feet for 20-30 seconds; land and disarm
Inspection	Disconnect battery; inspect frame and solder joints; feel motors/electronics for abnormal heat
Progression	Only repeat or extend flight if the first inspection is clean

Build Log, Safety, and Next Iteration

Record these values for every frame or electronics revision.

Frame revision / material	_____
Dry weight / battery weight / takeoff weight	_____
Motor / KV / propeller	_____
Battery / connector	_____
AIO firmware and Betaflight version	_____
Hover time and landing voltage	_____
Motor temperatures by touch or instrument	_____
Cracks, vibration, video noise, control issues	_____
One change for next revision	_____

LiPo and operating safety

- Use a quality balance charger and the correct LiPo cell count and charge current.
- Charge on a nonflammable surface and remain present. Store batteries appropriately; do not use damaged or swollen packs.
- Do not power a VTX without its antenna unless its manufacturer explicitly permits it.
- Keep propellers removed whenever the drone is connected to a computer or being serviced.
- For recreational flight in the United States, complete the FAA TRUST test and follow current FAA airspace and operating requirements.

Reference links

- Betaflight setup: <https://betaflight.com/docs/wiki/getting-started/setup-guide>
- Betaflight failsafe: <https://betaflight.com/docs/wiki/guides/current/Failsafe>
- ExpressLRS binding: <https://www.expresslrs.org/quick-start/binding/>
- FAA recreational flyers: https://www.faa.gov/uas/recreational_flyers
- RadioMaster Pocket: <https://radiomasterrc.com/products/pocket-radio-controller-m2>
- Example AIO specification: <https://www.speedybee.com/speedybee-f405-aio-40a-bluejay-25-5x25-5-3-6s-flight-controller/>

Next action

Print and measure the mock frame. Before buying the electronics, verify the exact frame STL, AIO listing, motor listing, motor-hole pattern, shaft/prop attachment, and expected component placement.